AUTOMOTIVE SOFTWARE SHOULD-COST · ALL-MODELS COMPARISON

Premium-SUV software cost — model-by-model

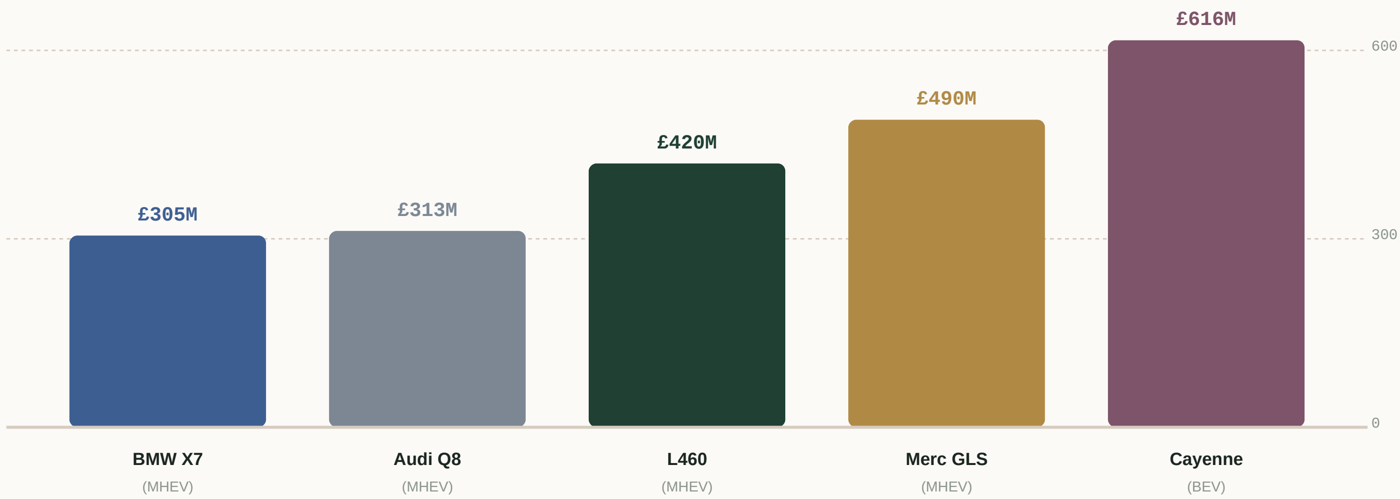
Every one of the 49 software modules priced across five flagship SUVs — Range Rover L460, BMW X7, Audi Q8, Mercedes GLS and Porsche Cayenne — side by side, apple-to-apple, with the cost drivers behind each difference.

MODELS COMPARED	MODULES EACH	WIDEST SPREAD
5	49	+58%

MARKET MAP

A Where the five models sit

Total software programme cost per model. The German MHEVs (X7, Q8, GLS) share the L460's mild-hybrid drivetrain; the Cayenne is shown on its full-BEV programme (49 vs 42 modules) and so carries the extra EV-powertrain stack. Reuse strategy, not badge, sets the order.



Reuse strategy — the real differentiator

BMW X7	`Heavy (0.35×)`	Lean — carries software forward across 7-Series / X5 / X7
Audi Q8	`Heavy + platform`	Lean — VW-Group shared middleware (0.14×)
Porsche Cayenne	`Medium + platform`	Mid — platform middleware, Porsche performance signature
Range Rover L460	`Medium (0.60×)`	High — develops more afresh; Tier-1 sourcing offsets
Mercedes GLS	`Medium (0.60×)`	Highest — Medium reuse + heavy MBUX signature + 9-yr life

Who leads

Across the 49 modules, BMW X7 is the leanest on 37 and Mercedes GLS the dearest on 42. The L460 is leanest on the 7 EV-powertrain modules (it beats the Cayenne there). The following pages price every module across all models.

A EV Powertrain & Battery

9 modules

BMS Core Software

spread £6.4M (15%)

Battery pack monitoring, protection logic, cell voltage/temp acquisition, state machine management, ASIL-D safety logic.

L460	£42.5M	LEANEST
Cayenne	£48.9M	DEAREST

Cayenne is dearest at £48.9M, L460 leanest at £42.5M — a £6.4M (15%) spread. Driven by baseline reuse and blended-rate differences (region / overhead / programme life).

EDU (Electric Drive Unit) Control Software

spread £4.8M (15%)

Integrated electric drive unit control, dual-motor torque vectoring, multi-speed gearbox integration, creep & one-pedal drive.

L460	£32.0M	LEANEST
Cayenne	£36.8M	DEAREST

Cayenne is dearest at £36.8M, L460 leanest at £32.0M — a £4.8M (15%) spread. Driven by baseline reuse and blended-rate differences (region / overhead / programme life).

Inverter Control Algorithms

spread £3.1M (15%)

Space Vector PWM, switching frequency optimisation, dead-time compensation, EMI management, demagnetisation protection.

L460	£20.5M	LEANEST
Cayenne	£23.6M	DEAREST

Cayenne is dearest at £23.6M, L460 leanest at £20.5M — a £3.1M (15%) spread. Driven by baseline reuse and blended-rate differences (region / overhead / programme life).

Motor Control (FOC/DTC/SVPWM)

spread £2.9M (15%)

Field-oriented control, direct torque control, sensorless rotor position estimation, flux linkage tables, temperature derating.

L460	£19.2M	LEANEST
Cayenne	£22.1M	DEAREST

Cayenne is dearest at £22.1M, L460 leanest at £19.2M — a £2.9M (15%) spread. Driven by baseline reuse and blended-rate differences (region / overhead / programme life).

SOC/SOH/SOE Estimation Models

spread £1.9M (14%)

Electrochemical & data-driven (ML) State of Charge, Health, Energy estimation. Kalman, EKF, neural network approaches.

L460	£13.6M	LEANEST
Cayenne	£15.5M	DEAREST

Cayenne is dearest at £15.5M, L460 leanest at £13.6M — a £1.9M (14%) spread. Driven by baseline reuse and blended-rate differences (region / overhead / programme life).

Fast-Charging Control Software

spread £1.1M (14%)

CCS/CHAdeMO/OCPP protocol stacks, dynamic power curve management, thermal derating during charge.

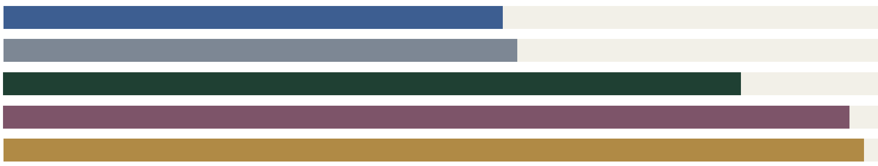
L460	£7.3M	LEANEST
Cayenne	£8.3M	DEAREST

Cayenne is dearest at £8.3M, L460 leanest at £7.3M — a £1.1M (14%) spread. Driven by baseline reuse and blended-rate differences (region / overhead / programme life).

Battery Thermal Management Software

spread £2.1M (72%)

Thermal control loops, coolant pump/valve actuation, fast-charge thermal preconditioning, cabin integration.



BMW X7	£3.0M	LEANEST
Audi Q8	£3.0M	
L460	£4.4M	
Cayenne	£5.0M	
Merc GLS	£5.1M	DEAREST

Merc GLS is dearest at £5.1M, BMW X7 leanest at £3.0M — a £2.1M (72%) spread. Driven by the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

Regenerative Braking Software

spread £2.0M (67%)

Brake blending control, hydraulic-electric transition, ABS/ESC coordination, one-pedal tuning, driver feel calibration.

Merc GLS is dearest at £4.9M, BMW X7 leanest at £3.0M — a £2.0M (67%) spread. Driven by the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

Cell Balancing Algorithms

spread £0.6M (15%)

Active/passive balancing algorithms, balancing current control, energy routing optimisation.

Cayenne is dearest at £4.8M, L460 leanest at £4.2M — a £0.6M (15%) spread. Driven by baseline reuse and blended-rate differences (region / overhead / programme life).

B ADAS L2 / L2+

9 modules

Camera Perception Stack

spread £16.6M (58%)

Object detection/classification (DNN), lane detection, traffic sign recognition, free-space estimation, parking vision. Mono + stereo cameras.

BMW X7	£28.7M	LEANEST
Audi Q8	£30.4M	
L460	£38.9M	
Cayenne	£43.5M	
Merc GLS	£45.3M	DEAREST

Merc GLS is dearest at £45.3M, BMW X7 leanest at £28.7M — a £16.6M (58%) spread. Driven by the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

Sensor Fusion Engine

spread £15.7M (66%)

Multi-modal object fusion (camera + radar + lidar if fitted), probabilistic world model, occupancy grid, track management.

BMW X7	£23.7M	LEANEST
Audi Q8	£24.5M	
L460	£33.7M	
Cayenne	£38.5M	
Merc GLS	£39.4M	DEAREST

Merc GLS is dearest at £39.4M, BMW X7 leanest at £23.7M — a £15.7M (66%) spread. Driven by the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

Path Planning Algorithms

spread £13.1M (67%)

Behaviour prediction, trajectory generation, cost-function optimisation, comfort/safety trade-off. HD map integration where applicable.

BMW X7	£19.6M	LEANEST
Audi Q8	£20.3M	
L460	£28.0M	
Cayenne	£32.0M	
Merc GLS	£32.7M	DEAREST

Merc GLS is dearest at £32.7M, BMW X7 leanest at £19.6M — a £13.1M (67%) spread. Driven by the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

Highway Assist / Traffic Jam Assist

spread £12.5M (64%)

Combined longitudinal + lateral control for highway driving, hands-off monitoring, lane change assist, automated speed adaptation.

BMW X7	£19.5M	LEANEST
Audi Q8	£20.2M	
L460	£27.4M	
Cayenne	£31.2M	
Merc GLS	£32.0M	DEAREST

Merc GLS is dearest at £32.0M, BMW X7 leanest at £19.5M — a £12.5M (64%) spread. Driven by the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

Control Algorithms (ACC / LKA / AEB)

spread £12.2M (63%)

Longitudinal ACC, lateral LKA, emergency AEB, FCW, emergency steering. Control law design, tuning, gain scheduling.

BMW X7	£19.5M	LEANEST
Audi Q8	£19.9M	
L460	£27.3M	
Cayenne	£31.4M	
Merc GLS	£31.7M	DEAREST

Merc GLS is dearest at £31.7M, BMW X7 leanest at £19.5M — a £12.2M (63%) spread. Driven by the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

Driver Monitoring Software (DMS)

spread £6.3M (62%)

Gaze tracking, drowsiness detection, attention estimation, hands-on-wheel detection. IR camera + CNN inference.

BMW X7	£10.2M	LEANEST
Audi Q8	£10.8M	
L460	£14.2M	
Cayenne	£16.0M	
Merc GLS	£16.5M	DEAREST

Merc GLS is dearest at £16.5M, BMW X7 leanest at £10.2M — a £6.3M (62%) spread. Driven by the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

Radar Processing Stack

spread £6.2M (68%)

FMCW radar DSP, CFAR detection, Doppler processing, angle estimation, object tracking and classification.

BMW X7	£9.2M	LEANEST
Audi Q8	£9.6M	
L460	£13.2M	
Cayenne	£15.1M	
Merc GLS	£15.4M	DEAREST

Merc GLS is dearest at £15.4M, BMW X7 leanest at £9.2M — a £6.2M (68%) spread. Driven by the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

Automated Parking & Surround-View

spread £3.6M (72%)

Automated parking assist (APA), remote/summon parking (RPA), 360° surround-view stitching, parking-slot detection, low-speed manoeuvre planning fusing

BMW X7	£5.0M	LEANEST
Audi Q8	£5.1M	
L460	£7.3M	
Cayenne	£8.4M	
Merc GLS	£8.6M	DEAREST

Merc GLS is dearest at £8.6M, BMW X7 leanest at £5.0M — a £3.6M (72%) spread. Driven by the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

Ultrasonic Processing Software

spread £0.8M (69%)

Short-range object detection, parking guidance, cross-traffic alert, time-of-flight processing for 12-16 sensors.

BMW X7	£1.2M	LEANEST
Audi Q8	£1.2M	
L460	£1.7M	
Cayenne	£2.0M	
Merc GLS	£2.0M	DEAREST

Merc GLS is dearest at £2.0M, BMW X7 leanest at £1.2M — a £0.8M (69%) spread. Driven by the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

C Infotainment & UX

9 modules

IVI Operating System (Android Automotive / QNX)

spread £9.7M (64%)

Android Automotive OS or QNX BSP integration, platform services, GPU driver optimisation, boot time optimisation, security hardening.

BMW X7	£15.0M	LEANEST
Audi Q8	£16.0M	
L460	£20.9M	
Cayenne	£23.5M	
Merc GLS	£24.7M	DEAREST

Merc GLS is dearest at £24.7M, BMW X7 leanest at £15.0M — a £9.7M (64%) spread. Driven by Merc GLS run it at Very-High complexity as a signature feature; the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

Vehicle Voice Assistant

spread £4.2M (42%)

Wake word detection, ASR (on-device + cloud), NLU, TTS, vehicle function control, 3rd-party assistant integration (Alexa/Google).

BMW X7	£10.0M	LEANEST
Audi Q8	£10.9M	
L460	£12.3M	
Cayenne	£13.3M	
Merc GLS	£14.2M	DEAREST

Merc GLS is dearest at £14.2M, BMW X7 leanest at £10.0M — a £4.2M (42%) spread. Driven by Merc GLS run it at Very-High complexity as a signature feature; the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

App Framework & HMI Layer

spread £4.2M (71%)

UI framework (Qt/Flutter/React Native Automotive), instrument cluster rendering, haptics, multi-screen management, UX design system.

BMW X7	£5.9M	LEANEST
Audi Q8	£6.2M	
L460	£8.5M	
Cayenne	£9.6M	
Merc GLS	£10.0M	DEAREST

Merc GLS is dearest at £10.0M, BMW X7 leanest at £5.9M — a £4.2M (71%) spread. Driven by the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

Navigation Stack

spread £2.9M (43%)

Routing engine, map rendering, offline capability, real-time traffic (HERE/TomTom), EV routing with charge planning.

BMW X7	£6.7M	LEANEST
Audi Q8	£7.4M	
L460	£7.6M	
Cayenne	£8.1M	
Merc GLS	£9.6M	DEAREST

Merc GLS is dearest at £9.6M, BMW X7 leanest at £6.7M — a £2.9M (43%) spread. Driven by Merc GLS run it at Very-High complexity as a signature feature; the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

Telematics Control Unit (TCU) Software

spread £2.4M (61%)

5G/LTE modem management, emergency call (eCall), remote diagnostics, remote access, V2X readiness, OBD-II data relay.

BMW X7	£4.0M	LEANEST
Audi Q8	£4.2M	
L460	£5.6M	
Cayenne	£6.3M	
Merc GLS	£6.5M	DEAREST

Merc GLS is dearest at £6.5M, BMW X7 leanest at £4.0M — a £2.4M (61%) spread. Driven by the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

AR Head-Up Display & AR Navigation

spread £1.4M (70%)

Augmented-reality head-up display compositor, sensor-registered navigation/ADAS overlays, optical distortion correction, graphics rendering pipeline for the

Merc GLS is dearest at £3.5M, BMW X7 leanest at £2.1M — a £1.4M (70%) spread. Driven by the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

Bluetooth / WiFi / 5G Connectivity Stack

spread £1.1M (58%)

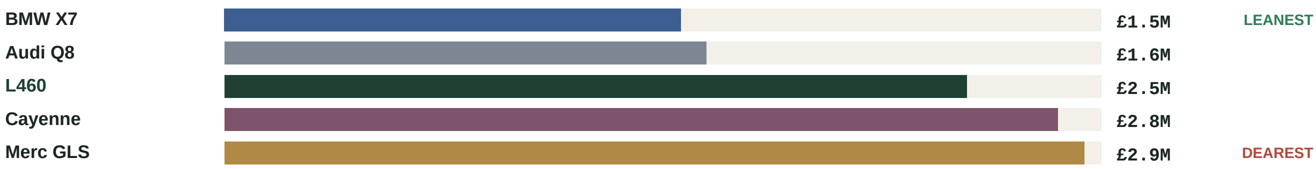
BT5.x stack (audio, phone), WiFi 6/6E AP+client, 5G SA/NSA modem driver integration, hotspot management.

Merc GLS is dearest at £2.9M, BMW X7 leanest at £1.9M — a £1.1M (58%) spread. Driven by the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

Premium Audio DSP & Active Noise Control

spread £1.3M (88%)

Branded 3D surround tuning (Meridian / Bowers & Wilkins / Bang & Olufsen / Burmester), active road/engine noise cancellation (ANC/RNC), active sound design,

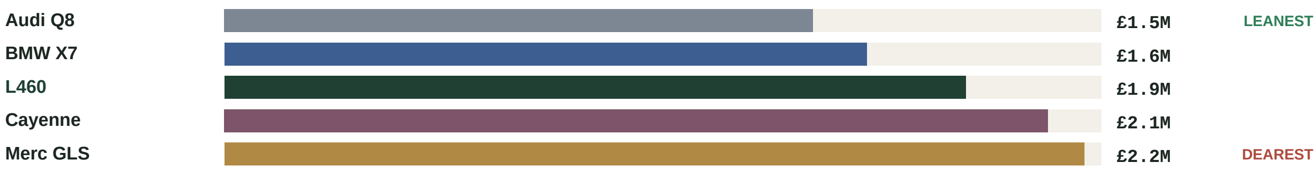


Merc GLS is dearest at £2.9M, BMW X7 leanest at £1.5M — a £1.3M (88%) spread. Driven by L460 / Cayenne / Merc GLS run it at Very-High complexity as a signature feature; the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

Digital Key (UWB/BLE) & Secure Access

spread £0.7M (46%)

CCC Digital Key 3.0 phone-as-key, UWB ranging & relay-attack protection, BLE fallback, secure-element / HSM integration, key sharing & cloud provisioning



Merc GLS is dearest at £2.2M, Audi Q8 leanest at £1.5M — a £0.7M (46%) spread. Driven by BMW X7 run it at Very-High complexity as a signature feature; the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

D Vehicle Domain Controllers

7 modules

Vehicle Motion Management (VMM)

spread £11.6M (69%)

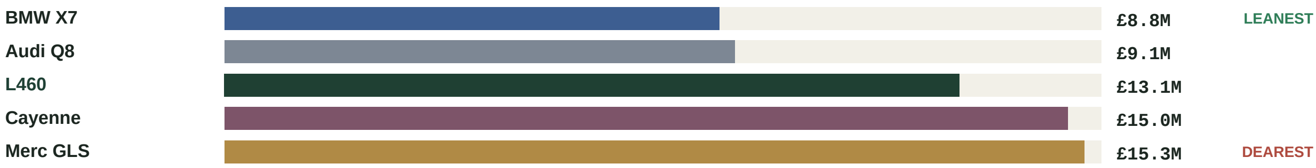
Unified torque arbitration, coordinated chassis & propulsion control, stability control, oversteer/understeer management, tyre model.

Merc GLS is dearest at £28.3M, BMW X7 leanest at £16.7M — a £11.6M (69%) spread. Driven by Cayenne run it at Very-High complexity as a signature feature; the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

Zonal Architecture Software

spread £6.5M (74%)

Zone controller software, power distribution management, ECU consolidation logic, 100BASE-T1 Ethernet backbone management.



Merc GLS is dearest at £15.3M, BMW X7 leanest at £8.8M — a £6.5M (74%) spread. Driven by the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

Chassis Control Software

spread £6.1M (67%)

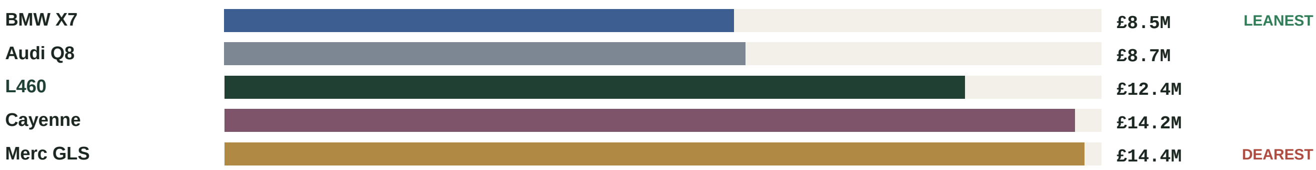
Air suspension, CDC (Continuous Damping Control), active anti-roll bars, 4WS (rear steering), hill-hold, terrain modes.

Merc GLS is dearest at £15.2M, BMW X7 leanest at £9.1M — a £6.1M (67%) spread. Driven by the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

Predictive Active Suspension & Body Control

spread £5.9M (69%)

Camera-fed road-surface preview, 48V active anti-roll / active body control (e.g. Mercedes E-Active Body Control, BMW Executive Drive Pro, JLR Dynamic

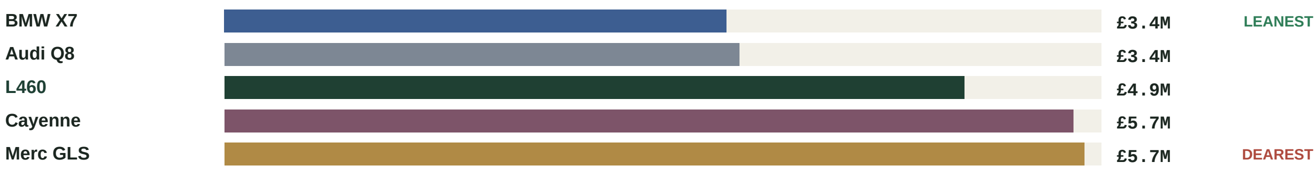


Merc GLS is dearest at £14.4M, BMW X7 leanest at £8.5M — a £5.9M (69%) spread. Driven by Cayenne / Merc GLS run it at Very-High complexity as a signature feature; the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

Body Control Software

spread £2.4M (71%)

Lighting control, power windows/mirrors/sunroof, door locks, wiper logic, ambient lighting, comfort features state machines.



Merc GLS is dearest at £5.7M, BMW X7 leanest at £3.4M — a £2.4M (71%) spread. Driven by the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

Gateway ECU Software

spread £2.0M (71%)

CAN/LIN/FlexRay/Ethernet routing, signal translation, diagnostic gateway, firewall, network management master.

Merc GLS is dearest at £4.9M, BMW X7 leanest at £2.9M — a £2.0M (71%) spread. Driven by the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

Climate & Thermal Comfort Control

spread £1.0M (71%)

Multi-zone HVAC control, heat-pump coordination, cabin air purification/ionisation, pre-conditioning, humidity & fogging management, seat/steering-wheel comfort

Merc GLS is dearest at £2.3M, BMW X7 leanest at £1.4M — a £1.0M (71%) spread. Driven by the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

E Middleware & Platform

6 modules

AUTOSAR Adaptive Platform

spread £8.0M (159%)

ara::com service-oriented communication, execution management, update management (UCM), PHM, crypto API, DDS integration.

Cayenne	£5.1M	LEANEST
Audi Q8	£5.3M	
BMW X7	£8.1M	
L460	£11.3M	
Merc GLS	£13.1M	DEAREST

Merc GLS is dearest at £13.1M, Cayenne leanest at £5.1M — a £8.0M (159%) spread. Driven by Cayenne / Audi Q8 share it as platform middleware (0.14× reuse).

AUTOSAR Classic BSW & Integration

spread £7.0M (138%)

AUTOSAR Classic BSW configuration (Vector DaVinci, ETAS ISOLAR), OS (OSEK/VDX), COM stack, run-time environment (RTE) integration.

Cayenne	£5.1M	LEANEST
Audi Q8	£5.3M	
BMW X7	£7.9M	
L460	£10.5M	
Merc GLS	£12.1M	DEAREST

Merc GLS is dearest at £12.1M, Cayenne leanest at £5.1M — a £7.0M (138%) spread. Driven by Cayenne / Audi Q8 share it as platform middleware (0.14× reuse).

CAN / LIN / FlexRay / Ethernet Communication Stacks

spread £1.7M (154%)

AUTOSAR ComStack configuration, CAN-XL support, Automotive Ethernet (10/100BASE-T1), signal routing, PDU mapping, SOME/IP.

Cayenne	£1.1M	LEANEST
Audi Q8	£1.2M	
BMW X7	£1.8M	
L460	£2.5M	
Merc GLS	£2.8M	DEAREST

Merc GLS is dearest at £2.8M, Cayenne leanest at £1.1M — a £1.7M (154%) spread. Driven by Cayenne / Audi Q8 share it as platform middleware (0.14× reuse).

Diagnostics (UDS / OBD-II)

spread £1.0M (60%)

UDS (ISO 14229), OBD-II (SAE J1979), DTC definition, diagnostic service implementation, DEXT generation, coding routines.

BMW X7	£1.6M	LEANEST
Audi Q8	£1.7M	
L460	£2.2M	
Cayenne	£2.5M	
Merc GLS	£2.6M	DEAREST

Merc GLS is dearest at £2.6M, BMW X7 leanest at £1.6M — a £1.0M (60%) spread. Driven by the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

RTOS & OS Porting

spread £1.4M (124%)

OSEK/VDX, FreeRTOS, or commercial RTOS BSP integration, task scheduling optimisation, interrupt management, memory protection.

Cayenne	£1.1M	LEANEST
Audi Q8	£1.2M	
BMW X7	£1.7M	
L460	£2.2M	
Merc GLS	£2.6M	DEAREST

Merc GLS is dearest at £2.6M, Cayenne leanest at £1.1M — a £1.4M (124%) spread. Driven by Cayenne / Audi Q8 share it as platform middleware (0.14× reuse).

Time Synchronisation & Network Management

`spread £0.4M (65%)`

gPTP (IEEE 802.1AS) for Ethernet time sync, CAN NM master, sleep/wake management, partial networking.

Merc GLS is dearest at £1.1M, BMW X7 leanest at £0.6M — a £0.4M (65%) spread. Driven by the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

F Cybersecurity (ISO 21434)

5 modules

Secure OTA Software Update Framework

spread £1.9M (51%)

Delta update generation, signature verification, rollback protection, update orchestration, bandwidth management.

BMW X7	£3.6M	LEANEST
Audi Q8	£3.9M	
L460	£4.7M	
Cayenne	£5.2M	
Merc GLS	£5.5M	DEAREST

Merc GLS is dearest at £5.5M, BMW X7 leanest at £3.6M — a £1.9M (51%) spread. Driven by the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

Secure Boot & Root of Trust

spread £1.7M (60%)

Hardware Security Module (HSM) integration, key provisioning, boot chain verification, anti-rollback, attestation.

BMW X7	£2.9M	LEANEST
Audi Q8	£3.0M	
L460	£4.0M	
Cayenne	£4.5M	
Merc GLS	£4.6M	DEAREST

Merc GLS is dearest at £4.6M, BMW X7 leanest at £2.9M — a £1.7M (60%) spread. Driven by the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

Intrusion Detection System (IDS)

spread £1.4M (49%)

In-vehicle network anomaly detection, CAN message monitoring, rate-limiting, VSOC integration, event reporting to cloud.

BMW X7	£2.8M	LEANEST
Audi Q8	£3.0M	
L460	£3.6M	
Cayenne	£4.0M	
Merc GLS	£4.2M	DEAREST

Merc GLS is dearest at £4.2M, BMW X7 leanest at £2.8M — a £1.4M (49%) spread. Driven by the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

Key Management System

spread £1.3M (51%)

Certificate lifecycle management, PKI integration, key derivation, secure key storage, provisioning infrastructure.

BMW X7	£2.5M	LEANEST
Audi Q8	£2.7M	
L460	£3.3M	
Cayenne	£3.7M	
Merc GLS	£3.8M	DEAREST

Merc GLS is dearest at £3.8M, BMW X7 leanest at £2.5M — a £1.3M (51%) spread. Driven by the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

Encryption Modules & Crypto Stack

spread £0.9M (65%)

AES-256, RSA-2048, ECC, TLS 1.3 for V2X/cloud, AUTOSAR Crypto Stack, hardware crypto acceleration.

BMW X7	£1.5M	LEANEST
Audi Q8	£1.5M	
L460	£2.1M	
Cayenne	£2.3M	
Merc GLS	£2.4M	DEAREST

Merc GLS is dearest at £2.4M, BMW X7 leanest at £1.5M — a £0.9M (65%) spread. Driven by the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

G OTA & Cloud Backend

4 modules

Cloud Backend Services

spread £5.2M (34%)

Vehicle connectivity backend, API gateway, device shadow, remote command, data lake, microservices architecture (AWS/Azure/GCP).

BMW X7	£15.6M	LEANEST
Audi Q8	£17.2M	
L460	£18.1M	
Cayenne	£19.1M	
Merc GLS	£20.8M	DEAREST

Merc GLS is dearest at £20.8M, BMW X7 leanest at £15.6M — a £5.2M (34%) spread. Driven by the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

OTA Update Manager (Vehicle-side)

spread £2.6M (51%)

Vehicle-side update campaign execution, ECU coordination, rollback, consent management, network condition handling.

BMW X7	£5.1M	LEANEST
Audi Q8	£5.4M	
L460	£6.7M	
Cayenne	£7.4M	
Merc GLS	£7.7M	DEAREST

Merc GLS is dearest at £7.7M, BMW X7 leanest at £5.1M — a £2.6M (51%) spread. Driven by the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

Data Pipeline & Telemetry

spread £1.6M (34%)

In-vehicle data collection agent, edge pre-processing, telemetry streaming, data lake ingestion, GDPR/data governance.

BMW X7	£4.7M	LEANEST
Audi Q8	£5.2M	
L460	£5.5M	
Cayenne	£5.8M	
Merc GLS	£6.3M	DEAREST

Merc GLS is dearest at £6.3M, BMW X7 leanest at £4.7M — a £1.6M (34%) spread. Driven by the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

Fleet Management & Analytics Software

spread £0.9M (32%)

Fleet health dashboard, predictive maintenance, usage analytics, dealer portal, subscription management, over-the-air campaign tools.

BMW X7	£2.9M	LEANEST
Audi Q8	£3.2M	
L460	£3.3M	
Cayenne	£3.5M	
Merc GLS	£3.8M	DEAREST

Merc GLS is dearest at £3.8M, BMW X7 leanest at £2.9M — a £0.9M (32%) spread. Driven by the leanest carry it forward (Heavy reuse) where L460 / Cayenne / Merc GLS develop more afresh (Medium).

LEAGUE TABLE

C Who is leanest, who is dearest

Count of modules on which each model is the leanest or the dearest of those that build it. A quick read on structural software-cost position.

MODEL	MODULES LEANEST	MODULES DEAREST
BMW X7	37	0
Audi Q8	1	0
L460	7	0
Cayenne	4	7
Merc GLS	0	42

BMW X7 leads on cost across the board (Heavy reuse + an 8-year programme). Audi Q8 is close but its 9-year life adds lifecycle cost. Mercedes GLS is dearest almost everywhere (Medium reuse, Very-High MBUX signature, 9-year life). The L460 and Cayenne sit between — the L460's Medium reuse is its one structural gap, the Cayenne's cost is its BEV scope and Porsche performance software.

TRANSPARENCY

D Assumptions & methodology

Every figure is live output from the CostVision should-cost engine (49-module catalogue, ISO 26262 / 21434). Nothing is hand-entered.

Comparison basis

Each model on its real drivetrain: the German MHEVs (X7, Q8, GLS) and the L460 on 48V mild-hybrid; the Cayenne on its full-BEV programme. Non-powertrain module cost is drivetrain-independent, so is directly comparable; the 7 EV-powertrain modules exist only on the L460 (BEV) and Cayenne.

Per-module cost

Total programme should-cost (development + testing + integration + cybersecurity + calibration + toolchain, plus maintenance + cloud + IP over programme life).

What sets the spread

Reuse strategy (Fresh 1.00 / Light 0.82 / Medium 0.60 / Heavy 0.35 / Platform 0.14 × dev effort), signature-software complexity, region / dev-source blended rate, overhead and programme life.

Model assumptions

L460: UK · Tier-1 · 75k · 8yr · Medium. X7: EU · in-house · 60k · 8yr · Heavy. Q8: EU · in-house · 55k · 9yr · Heavy + platform. GLS: EU · in-house · 45k · 9yr · Medium. Cayenne: EU · in-house · 50k · 8yr · Medium + platform + Porsche performance.

Accuracy

Bottom-up parametric should-cost; validated envelope ±25–35% (back-test MAPE 24.9%). Absolute numbers are estimates; comparisons are consistent (same engine + rate library).

Propulsion availability

Verified July 2026: X7 / Q8 / GLS are MHEV-only in production; Cayenne offers ICE/PHEV/BEV; Range Rover Electric is launching. No fabricated variants are compared.

Currency & scope

All figures GBP. Embedded & connected vehicle software only; excludes manufacturing, hardware and non-software engineering.

This report supports target-setting and supplier negotiation; it is not an audited actual. Engine: CostVision sw-should-cost.ts, rate library v1.0, July 2026.